American CRR Pricing versus the European Benchmark

CHEME 5660
Cornell University
October 22, 2026
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LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

- **Contrast exercise styles.** Price European and American contracts on the same lattice and explain why the American value is never smaller under identical inputs.
- **Locate exercise decisions.** Use backward induction to identify the nodes where early exercise makes the American premium strictly larger.
- **Separate value components.** Distinguish intrinsic value, continuation value, extrinsic value, and the early-exercise premium without conflating their meanings.

1 An American Option Contains a Timing Choice

An American option is not merely a terminal payoff. It is a contract that lets its holder choose an exercise time on or before expiration. That flexibility converts valuation into an optimal-stopping problem: at every admissible decision time, compare the cash flow from exercising now with the value of retaining the contract.

Let $S_{t,k} > 0$ be the underlying price at node (t, k) of a recombining lattice, let $K > 0$ be the strike in USD/share, and let $T > 0$ be expiration. The intrinsic-value functions are

$$h_c(S) := (S - K)^+, \qquad h_p(S) := (K - S)^+, \qquad x^+ := \max\{x, 0\}. \tag{1}$$

Intrinsic value is the cash flow from immediate exercise; it is not the option's full economic value before expiration.

For N steps of length $\Delta t := T/N$, the Cox–Ross–Rubinstein (CRR) factors and risk-neutral up probability for a non-dividend-paying underlying are

$$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = e^{-\sigma\sqrt{\Delta t}}, \qquad q = \frac{e^{g_y \Delta t} - d}{u - d}. \tag{2}$$

The step must satisfy $d < e^{g_y \Delta t} < u$, equivalently $0 < q < 1$. The weight q is a pricing probability rather than a forecast of the real-world frequency of up moves. This is the one-step risk-neutral probability of L3b, written for a continuously compounded rate: setting the risk-free gross return $R_f = e^{g_y \Delta t}$ recovers $q = (R_f - d)/(u - d)$ exactly. We keep L3bs letters, where p is the real-world up probability and q the pricing weight, and L3bs rate symbol r_f becomes the g_y of L2a and L9a. Equation Eq. 2 assumes no dividend; with a continuous dividend yield δ the underlying drifts at $g_y - \delta$ under the pricing measure and the probability becomes $q = (e^{(g_y-\delta)\Delta t} - d)/(u - d)$.

For a step length $\Delta t > 0$, continuously compounded risk-free rate $g_y \in \mathbb{R}$, and risk-neutral up probability $q \in (0, 1)$, define the discounted continuation value

$$C_{t,k} := e^{-g_y \Delta t} \left[q V_{t+1,k+1} + (1 - q) V_{t+1,k} \right]. \tag{3}$$

Here $V_{t+1,k+1}$ and $V_{t+1,k}$ are the optimally managed American-option values at the two child nodes. The Bellman recursion is

$$V_{t,k} = \max\{h(S_{t,k}), C_{t,k}\}, \qquad V_{N,k} = h(S_{N,k}), \tag{4}$$

where h is the appropriate function from Eq. 1. Exercise is strictly optimal when $h > C$; continuation is strictly optimal when $C > h$. Equality is an indifference node: both actions carry the same value, so the premium does not depend on the convention, but the reported decision does. The course package takes the maximum of the two and the L9b example notebook reports exercise on a tie.

For an otherwise identical European contract, terminal values are the same but every earlier node must continue:

$$V^{\mathrm{EU}}_{t,k} = e^{-g_y \Delta t}\left[qV^{\mathrm{EU}}_{t+1,k+1} + (1-q)V^{\mathrm{EU}}_{t+1,k}\right]. \tag{5}$$

The American strategy set contains the European strategy of waiting until expiration. Therefore

$$V_0^{\mathrm{AM}} \geq V_0^{\mathrm{EU}}. \tag{6}$$

The inequality is strict only when an early-exercise opportunity changes the root value; American style alone does not guarantee a higher premium.

Theorem 1.1. Value decomposition at an American-option node

Define extrinsic value at an interior node (t,k), that is a node with $t < N$, by

$$X_{t,k} := V_{t,k} - h(S_{t,k}). \tag{7}$$

Then the Bellman recursion Eq. 4 and the identity $\max\{h, C\} - h = \max\{0, C - h\}$ give

$$X_{t,k} = \left(C_{t,k} - h(S_{t,k})\right)^{+} \geq 0. \tag{8}$$

Continuation value is not defined at expiration, so the terminal case is separate: $V_{N,k} = h(S_{N,k})$ gives $X_{N,k} = 0$ directly. Thus $X = 0$ at expiration, throughout the exercise region, and at any indifference node where $h = C$, while $X = C - H > 0$ in the strict continuation region.

Remark 1.1. Three quantities that sound similar

Continuation value C is the value of waiting. Extrinsic value $X = V - h$ is the amount by which the optimally managed contract exceeds immediate exercise value. The early-exercise premium $A := V^{\mathrm{AM}} - V^{\mathrm{EU}}$ is the value of American exercise flexibility relative to an otherwise identical European contract. In general, C, X, and A are different quantities.

2 A Two-Step American Put

Consider the textbook put example of Hull Examples 13.4 and 13.5, with $S_0 = 50$ USD/share, $K = 52$ USD/share, two one-year steps, $u = 1.2$, $d = 0.8$, and $g_y = 0.05\ \mathrm{yr}^{-1}$, so that Eq. 2 gives $q \approx 0.6282$. These rounded factors satisfy $ud = 0.96 \neq 1$, so this is a generic binomial lattice rather than a CRR calibration and no single volatility is implied by them; the recursions of Eq. 4 and Eq. 5 are unchanged either way. See Fig. 1. At the one-year down node, $S = 40$ and immediate exercise pays $h_p(40) = 12$. Waiting is worth

$$C_{1,0} = e^{-0.05}\left[0.6282(4) + 0.3718(20)\right] \approx 9.46, \tag{9}$$

so exercise is optimal and $V_{1,0} = 12$. At the one-year up node, $S = 60$, intrinsic value is zero, and continuation is

$$C_{1,1} = e^{-0.05}\,[0.6282(0) + 0.3718(4)] \approx 1.41, \tag{10}$$

so the holder waits. Backward induction gives an American root value of approximately 5.09 USD/share, versus 4.19 USD/share for the European put. At the root, intrinsic value is 2, extrinsic value is about 3.09, and the early-exercise premium is about 0.90. These three numbers are a numerical demonstration of the distinctions in Remark 1.1.

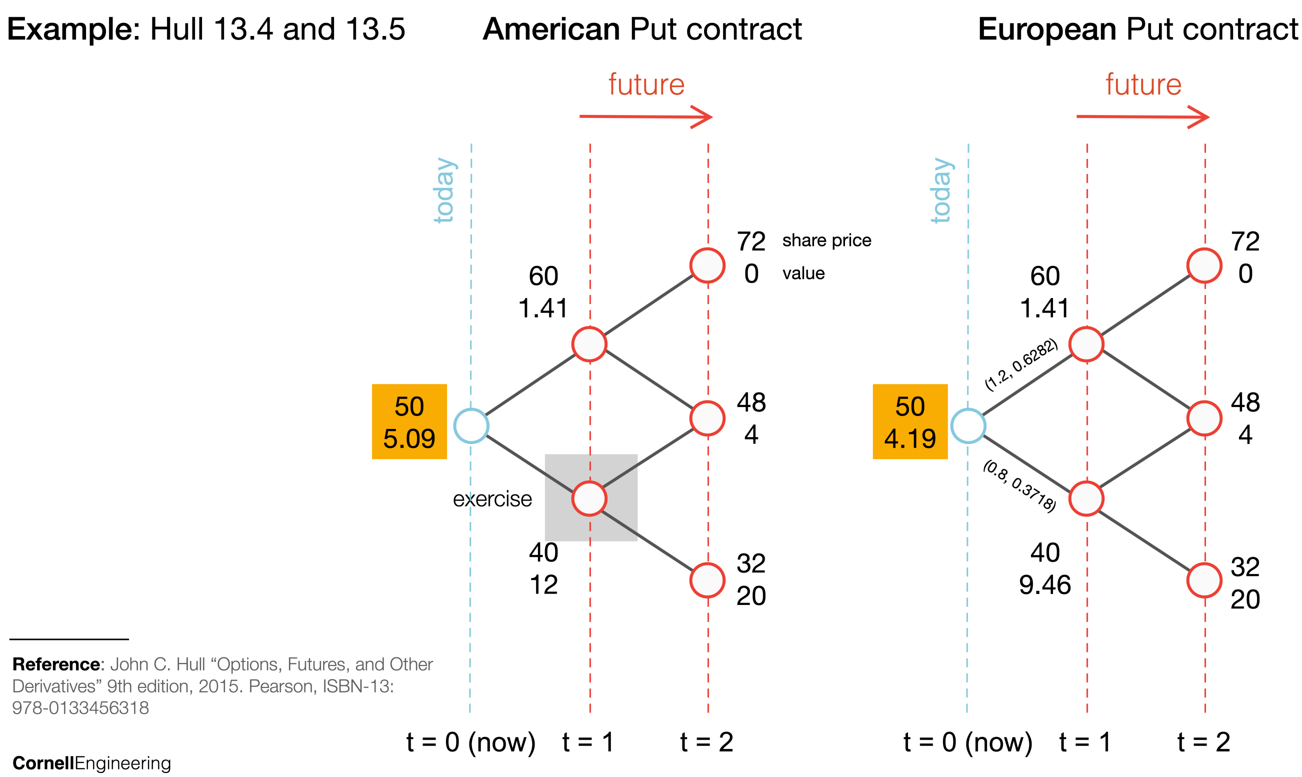

Figure 1: American and European put lattices for the two-step example. The American holder exercises at the $S = 40$ node because $12 > 9.46$; the European holder must continue. Values are in USD/share.

COMPANION NOTEBOOK
L9b: American versus European Put Example →
Price both exercise styles on the same two-step lattice, measure the early-exercise premium, and inspect the node where exercise dominates continuation.

3 Moneyness Is Not the Same as Value

Moneyness describes the relation between spot and strike. A call is in the money (ITM) when $S > K$, at the money (ATM) when $S \approx K$, and out of the money (OTM) when $S < K$; the inequalities reverse for a put. Moneyness determines intrinsic value through Eq. 1, but it does not determine the entire premium.

An OTM option has $h = 0$ yet can have $V > 0$ because future states with positive payoff remain possible. In that case the entire model value is extrinsic: $X = V$. For an ITM option, $h > 0$, but waiting can still be valuable because it preserves favorable upside states, avoids paying the strike early for a call, or delays surrendering the underlying for a put.

Fig. 2 sketches the decomposition. For fixed model inputs, extrinsic value is often largest near the exercise boundary or near the money. Far OTM options have little probability-weighted payoff; deep ITM options increasingly resemble immediate exercise or a linear position. These are qualitative tendencies, not universal monotonic laws.

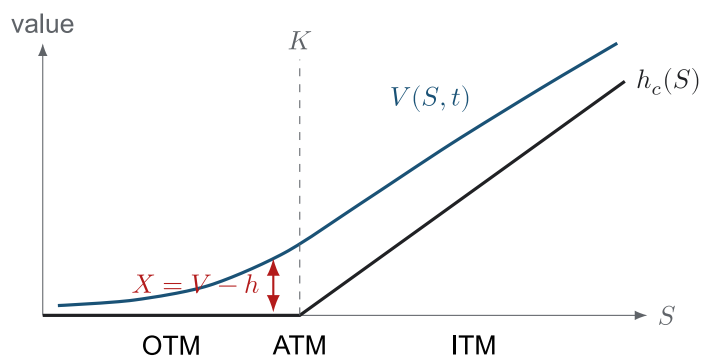

Figure 2: Schematic call-value decomposition before expiration. The blue premium curve lies on or above intrinsic value in an arbitrage-free American model, touching it wherever exercise is optimal; their vertical gap is extrinsic value. The exact shape depends on time, volatility, rates, dividends, and exercise policy.

COMPANION NOTEBOOK

L9b: NVDA Intrinsic and Extrinsic Value →

Decompose modeled call premiums across strike and time to expiration, and connect the premium–intrinsic gap to moneyness.

4 What Moves Extrinsic Value?

Extrinsic value reflects the option's remaining opportunity set under the pricing model. Holding other inputs fixed, more time and higher volatility often increase a vanilla option's opportunity to finish favorably. Interest rates affect the value of deferring the strike payment or receipt, while dividends change the economics of owning the stock rather than the option. These effects interact with contract type and the endogenous exercise boundary.

The phrase *time decay* must be used carefully. Along a fixed hypothetical state with all other inputs held constant, less remaining time generally removes optionality. Along an observed market path, however, S, implied volatility, rates, dividends, liquidity, and the exercise boundary all move. Consequently, a sequence of quoted extrinsic values need not decline smoothly. Theta, introduced in L10a, is a local model derivative; it is not a promise that tomorrow's market premium will be lower.

For a non-dividend-paying stock and $g_y \geq 0$, early exercise of an American call is never strictly better in the classical frictionless model, so $V_c^{\text{AM}} = V_c^{\text{EU}}$. Waiting is strictly better at every node where the contract still has positive value; where it is worthless both actions are worth zero, and at $g_y = 0$ the holder can be indifferent even in the money. In every case the two premiums are equal. American puts can optimally exercise early because receiving K sooner has financing value, but that benefit must exceed the optionality surrendered, which is why the put also has to be sufficiently in the money. Dividends can create early-exercise incentives for calls, and negative rates or trading constraints can change the standard conclusions.

5 From Model Values to an Option Chain

For a quoted market premium P^{mkt}, a descriptive decomposition uses

$$X^{\text{mkt}} := P^{\text{mkt}} - h(S). \tag{11}$$

The choice of bid, ask, last trade, or midpoint matters. Wide spreads, stale prints, asynchronous spot prices, discrete dividends, and contract multipliers can distort comparisons. A negative value from Eq. 11 is usually a data or executability warning, not evidence of negative model extrinsic value.

Likewise, a CRR price outside a quoted bid–ask spread does not by itself prove a tradable mispricing. First reconcile timestamp, spot, exercise convention, dividend assumptions, rate curve, implied volatility, tree resolution, and transaction costs. Model disagreement is a diagnostic signal; arbitrage is a stronger claim requiring an executable strategy and consistent inputs.

> **COMPANION NOTEBOOK**
> **L9b: American Contract Premiums versus AMD Quotes →**
> Price American calls and puts across strikes with CRR and compare the model values with observed bid–ask intervals.

Optional Advanced Material

The lattice factors in Eq. 2 are quoted rather than derived. The following notebook is optional and is not a prerequisite for L10a.

> **COMPANION NOTEBOOK**
> **L9b Advanced: Where the CRR Factors Come From →**
> Derive the reciprocal CRR factors of Eq. 2 by matching the variance of a two-point lattice log return to the diffusion variance, separate the reciprocal condition $ud = 1$ as a modeling convention rather than a consequence of the diffusion, state the order of the approximation and the term it discards, and give the step sizes for which $0 < q < 1$.

KEY TAKEAWAYS

In this lecture, we priced European and American exercise styles on the same lattice, measured the early-exercise premium, and connected the difference to node-level decisions.

- **Extrinsic value is a residual.** $X = V - h = (C - h)^{+}$ at an interior lattice node, with $X = 0$ at expiration; continuation value, extrinsic value, and the early-exercise premium are not synonyms.
- **American value weakly dominates European value.** The American holder can always wait until expiration, so $V^{\mathrm{AM}} \geq V^{\mathrm{EU}}$; strict inequality requires a valuable early-exercise opportunity.
- **Quoted decompositions require care.** Bid, ask, timestamp, dividends, volatility, and liquidity must be aligned before treating model–market differences as economic opportunities.

Next, measure how an option premium responds locally to spot, volatility, time, and interest rates through the Greeks.

ADDITIONAL READING

- John C. Cox, Stephen A. Ross, and Mark Rubinstein, "Option Pricing: A Simplified Approach," *Journal of Financial Economics* 7(3), 229–263 (1979).
- Robert C. Merton, "Theory of Rational Option Pricing," *Bell Journal of Economics and Management Science* 4(1), 141–183 (1973).
- John C. Hull, *Options, Futures, and Other Derivatives*, 9th ed., Pearson (2015), ISBN 978-0133456318, Chapter 13 for binomial trees and American options.